

Python for QA - Installation & Setup

A complete guide to prepare the environment, install, configure, validate and perform the first functional use of the tool.

Manual objective

This manual guides you through the complete installation and setup process, from preparing the environment to final validation. Follow the steps in order and use the checks throughout the guide to confirm that each configuration was completed correctly.

Official references

<https://www.python.org/downloads/>

Virtual environment

Use `python -m venv .venv` to isolate dependencies. In PowerShell activate with `.\venv\Scripts\Activate.ps1`.

How to start safely

Start from the official Python for QA site and documentation. Confirm prerequisites, supported versions, operating system and required permissions. Learn in a controlled environment before using it in a critical workflow.

First QA-oriented use

Pick one small, verifiable task in Python for QA, configure only what is necessary, run a simple example and confirm the outcome. Repeat with different data or conditions before adding abstractions.

Common issues and troubleshooting

When Python for QA fails, verify versions, dependencies, permissions, connectivity, data and environment before assuming a product defect. Preserve the original error and reduce the issue to the smallest reproducible case.

Installation overview

Use the official Python installer for Windows or the mechanism recommended by current Python documentation. Pay attention to options that make Python available in the terminal. Open a new terminal and validate `python --version` and `py --version`.

pip and virtual environment

Validate packages with `python -m pip --version`. For projects prefer a virtual environment: `python -m venv .venv`, then activate it before installing dependencies.

Forgot PATH

Try `py --version` first. If Python is installed but `python` is not recognized, locate the installation and add the directory containing `python.exe` and, when needed, `Scripts` to Path. Reopen the terminal.

First script

Create `hello.py` with `print("QA Lab Brasil")` and run `python hello.py`; `py hello.py` may also be used on Windows.

Troubleshooting

If pip targets another Python, prefer `python -m pip`. With multiple versions use where `python` and `py -0p` when available.